Blockers:
Unitary evolution as infinitesimal projective processes

This file collects everything about the two brief issues of `open-problems.tex` that block verification of Theorem 18, together with a proposed resolution of each and the changes it needs in the brief. Sections 1 and 2 are copied from `open-problems.tex` and `formalization-log.tex`; the rest is new.

1 The blockers

From `open-problems.tex`, "From the formalization":

1. [BLOCKS VERIFICATION: BRIEF ISSUE] **Durations with irrational ratio.** DR-PROJ defines Φ_t only as $\lim_n(\Psi_{t/n})^n$. For t/s irrational a common step τ gives blocks Ψ_τ^k that are not of this form, so "common refinements" and continuity at zero do not prove $\Phi_{t+s} = \Phi_t \circ \Phi_s$; eq. (26) would have to hold uniformly along the trajectory. The remaining interval itself is harmless. A fix: in finite dimension a telescoping estimate gives $\Phi_t = e^{tL}$ directly, which is formalized (`IsProjectiveLimit.exists_eq_exp`); in infinite dimension Chernoff's product formula does, if the closure of L generates a contraction semigroup, which is open. In infinite dimension this is the one clause of `IsEvolution` that is assumed; the others, including composition for durations of rational ratio (eq. (29)), are derived (`IsProjectiveLimit.isEvolution`). $\Phi_0 = I$ is assumed in both dimensions, since the formalization defines Φ_t only for $t > 0$; the brief obtains it from $\Psi_0 = I$.

   The gap may be real in infinite dimension. A sketch, not checked in Lean: let $H_1 \neq H_2$ be two self-adjoint extensions of a symmetric operator H_0 with deficiency indices $(1, 1)$, and let Ψ_τ be the cutoff resolvent of H_1 for rational τ and of H_2 for irrational τ. Since t/n is rational exactly when t is, limits, prediction and reconstruction hold for each t, and Φ_t is conjugation by e^{-itH_1} or e^{-itH_2}. If both families have the derivative $[H_0, \rho]/(i\hbar)$ on the density operators built from vectors of $\mathcal{D}(H_0)$, which are dense, then DR-PROJ holds while DR-UNIT would force $H_1 = H_2 + c$. (Brief, Sec. 4, after eq. (29); Log, "Problems found".)

2. [BLOCKS VERIFICATION: BRIEF ISSUE] **What perfect reconstruction means.** DR-PROJ only asks that the probabilities of the reconstructed sequences tend to one. The formalization instead assumes a second projective limit Φ'_t with $\Phi'_t(\Phi_t P) = P$ and $\Phi_t(\Phi'_t P) = P$ on pure states. It is not settled whether the brief's weaker wording already makes Φ_t a bijection of the pure states, or what the right formulation is. (Brief, Sec. 4: "Perfect reconstruction gives the converse correspondence"; Log, "Differences" and "Problems found".)

2 Related items

From `open-problems.tex`, "From the formalization" (the preamble, which places the blockers in the any-dimension formalization):

> Beyond finite dimension the formalization has a part in any dimension, which covers unbounded H in DR-UNIT $\Rightarrow$ DR-PROJ and derives the evolution structure `IsEvolution` from DR-PROJ up to the item "Durations with irrational ratio" below (`drUnit_iff_drProj_and_map_zero_and_map_add`), for every cutoff level satisfying eq. (22).

The former item "The generator of the cutoff family" of `open-problems.tex`, needed if the first blocker is resolved through the generator by Chernoff's formula, is resolved. On the core, the generator of the family (23) is $\mathcal{L}(\rho) = [H, \rho]/(i\hbar)$: ρ maps into the domain of H, and $\mathcal{L}(\rho)\psi = -i(H\rho\psi - \rho H\psi)$ for ψ in the domain (`cutoffResolvent_generator`; Log, Sec. 1.1, "Differences", item on eq. (26)).
The former item "Complete positivity of the resolvent of an unbounded generator" of `open-problems.tex`, from the brief, is resolved. A matrix of operators is positive exactly when its block operator on E^k is (`cstarMatrix_nonneg_iff`), so complete positivity is a family of pointwise inequalities, which the strongly convergent Laplace integral keeps (`exists_completelyPositiveMap_resolvent`; Log, Sec. 1.1, "Differences", last item).
From `formalization-log.tex`, Section 1, "Differences from the brief":

- DR-PROJ is made precise as follows. Outcomes and branches refer to a projective decomposition that realizes Ψ on the current pure state. Perfect prediction asks that the final branch state converges, without naming the limit; DR-PROJ $\Rightarrow$ DR-UNIT shows that it is $\Phi_t P$. Reconstruction is a second projective limit that inverts Φ_t on pure states. For the resolvents it is the limit of $(1 + \tau L)^{-1}$, as in the remark "The finite process is not unitary".

- `drUnit_iff_drProj` assumes `IsEvolution` of the brief on the Schrödinger equation: Φ_t is affine and preserves density operators, $\Phi_0 = I$, $\Phi_{s+t} = \Phi_s \Phi_t$, and $t \mapsto \Phi_t \rho$ is continuous at 0^+. The brief derives these from DR-PROJ (affinity of the limit, eqs. (28)–(29), common refinements). `IsProjectiveLimit.isEvolution` derives them too, but not by the brief's route. Each Ψ_τ agrees on density operators with a linear map T_τ, because the density operators contain a basis of the operators; $(T_\tau - I)/\tau \to B$ in operator norm, because convergence on a basis is norm convergence; and $(T_{t/n})^n \to e^{tB}$ by telescoping. So $\Phi_t = e^{tB}$ on density operators for every $t > 0$, which gives affinity, preservation of density operators, continuity at 0^+ and composition for all durations at once; eq. (28) and the common refinements are not needed. DR-PROJ says nothing about Φ_0, which is assumed to be the identity: `drUnit_iff_drProj_and_map_zero` is DR-UNIT $\Leftrightarrow$ DR-PROJ with $\Phi_0 = I$.

From `formalization-log.tex`, Section 1, "Problems found in the brief":

- Section 3 states DR-PROJ only as "the probabilities of the predicted and reconstructed state sequences tend to one". The remark on prediction and reconstruction adds that the predicted states converge. This is not an extra assumption: since the branch is within $2(1 - w)$ of $\Psi^n P \to \Phi_t P$, the predicted states converge to $\Phi_t P$ anyway. The finite-dimensional proof still uses the clause; the proof in any dimension does not (Section 1.1).

- "Perfect reconstruction gives the converse correspondence" does not say how the reconstructed sequence relates to the predicted one. Injectivity and surjectivity of Φ_t on pure states need the reconstruction to invert Φ_t; this is what DR-PROJ assumes here.

- Composition for durations with irrational ratio, "take common refinements with a remaining interval whose duration tends to zero", is a gap. The remaining interval is harmless: with a common step τ the powers of Ψ_τ commute, so $\|\Psi_\tau^k(\Psi_\tau^j\rho-\rho)\|_1 \le j\|\Psi_\tau\rho-\rho\|_1 \approx j\tau\|L\rho\|_1 \to 0$ by the telescoping of eq. (28). The problem is the step size. DR-PROJ defines Φ_t only as $\lim_n(\Psi_{t/n})^n$. With t/s irrational and $\tau = (t+s)/N$, the block of $k = \lfloor t/\tau \rfloor$ steps is Ψ_τ^k, which is not of the form $(\Psi_{t/k})^k$, and DR-PROJ says nothing about its limit. Comparing $(\Psi_a)^k$ with $(\Psi_b)^k$ for nearby a, b needs the $o(\tau)$ in eq. (26) to be uniform along the states $\Psi_b^j\rho$, while eq. (26) holds for each ρ separately. Continuity at zero does not supply this. In finite dimension L is bounded, eq. (26) holds in operator norm, and telescoping gives $\|(\Psi_{t/n})^n - e^{tL}\| \le n\, o(t/n)\, e^{t\|L\|} \to 0$, so $\Phi_t = e^{tL}$ and composition follow at once (formalized, `IsProjectiveLimit.exists_eq_exp`). In infinite dimension Chernoff's product formula gives the same, uniformly on compact t-intervals, provided the closure of L generates a contraction semigroup; the brief's "single, possibly unbounded generator L" must be read to include this, or it must be assumed.

From `formalization-log.tex`, Section 1.1 (any dimension), "Differences from the brief":

- DR-PROJ $\Rightarrow$ DR-UNIT is proved over `IsEvolution` (`DrProj.drUnit`) and over two of its clauses only (`drUnit_iff_drProj_and_map_zero_and_map_add`). The finite-dimensional derivation of `IsEvolution`, through $\Phi_t = e^{tB}$, needs a finite basis of density operators; `IsProjectiveLimit.isEvolution` follows the brief instead. Each Φ_t is a trace-norm limit of affine maps that keep density operators, so it is one too, and contracts the trace norm (Simon's lemma, the brief's "extend to positive trace-preserving linear maps"). Eq. (28) is the telescoping bound $\|\Psi^n\rho - \rho\|_1 \le n\|\Psi\rho - \rho\|_1$ on the dense set of eq. (26), and contractivity extends continuity at 0^+ to every density operator. Eq. (29) is proved as $\Phi_{(p+q)u} = \Phi_{pu} \circ \Phi_{qu}$, along $n = (p+q)m$, which gives composition whenever s/t is rational. Two clauses remain assumptions: $\Phi_0 = I$, since `IsProjectiveLimit` defines Φ_t only for $t > 0$ (the brief obtains it from $\Psi_0 = I$), and $\Phi_{s+t} = \Phi_s \circ \Phi_t$ for s/t irrational, the gap noted in the finite-dimensional part.
- The proof of DR-PROJ $\Rightarrow$ DR-UNIT does not use the clause of `PerfectPrediction` that the final branch states converge. The bound $\|\Psi^n P - Q_n\|_1 \le 2(1-w_n)$ and $\Psi^n P \to \Phi_t P$ already give $Q_n \to \Phi_t P$. The clause is kept so that the definitions match those of finite dimension.

From `formalization-log.tex`, Section 1.1, "Problems found in the brief":

- Composition for durations of irrational ratio is the gap noted in the finite-dimensional part. Here no telescoping estimate closes it: L may be unbounded, and eq. (26) holds only for each ρ of a dense set, not uniformly.

3 The author's proposal (2026-11-02)

Paraphrased, from a reply on another brief that the author means for all briefs: the briefs get a section of mathematical prerequisites, which states what Lean should prove and is fully formalized. The relations between physical conditions, such as Theorem 18, are argued informally from these prerequisites and are checked by reading, not by Lean. The author will rewrite the briefs accordingly.

This brief already has such a section (Sec. 2: Definition 1, Propositions 2, 13 and 17), and it is formalized. Both blockers sit in Theorem 18, in what DR-PROJ means. Neither is a question of reading a physical condition charitably: each decides which mathematical statement the proof of Theorem 18 rests on. So DR-PROJ has to be stated so that the proof goes through, and the facts the proof then uses belong in the prerequisites.

4 Proposed resolution

4.1 Durations with irrational ratio

Guess. DR-PROJ means that the evolution does not depend on how the elapsed time is subdivided. The limit is taken over *every* subdivision of $[0, t]$ whose longest interval tends to zero, not only over the equal subdivisions t/n:

$$\Phi_t \rho = \lim \Psi_{\tau_k} \circ \cdots \circ \Psi_{\tau_1} \rho \qquad \text{in trace norm, for } \tau_i > 0,\ \textstyle\sum_i \tau_i = t,\ \max_i \tau_i \to 0,$$

and $\Phi_0 = I$, the empty product. Composition is then immediate (prerequisite P1): a subdivision of $[0, s]$ followed by one of $[0, t]$ is a subdivision of $[0, s + t]$. DR-UNIT still implies the stronger DR-PROJ (P2), by the telescoping argument of the brief, run on the core of bounded spectral support. The equal subdivisions are a special case, so every other step of the proof stands as it is. The counterexample sketch of Section 1 is excluded: a subdivision of a rational t into irrational steps leads to e^{-itH_2}, not e^{-itH_1}.

Alternative. Keep the equal subdivisions and assume instead that the closure of the generator L of eq. (26) generates a contraction semigroup. Chernoff's product formula then gives $\Phi_t = e^{t\bar{L}}$, and composition follows. For the direction DR-UNIT ⇒ DR-PROJ this needs $\bar{L}$ to be the generator of $\rho \mapsto U_t \rho U_t^\dagger$; on the core, L is $[H, \cdot]/(i\hbar)$ (`cutoffResolvent_generator`), but that the core is a core for this generator is not formalized, and Chernoff's formula is not in Mathlib.

Why.

- It is what the brief already says. "The use of the same duration-dependent process makes the limiting evolution depend only on elapsed time" and "Composition follows by taking common refinements" make sense only if unequal steps are allowed: a common refinement of t and s is not an equal subdivision of $t + s$ when t/s is irrational.
- Time is homogeneous. An observer who checks the system at uneven times, say on a clock that ticks irregularly, should see the same evolution as one who checks at even times. A process for which the outcome depends on whether the step lengths are rational is not a physical process.
- It costs no physics. The direction DR-UNIT ⇒ DR-PROJ survives with the same estimates (P2), so Theorem 18 keeps its full range, including unbounded generators.
- It is elementary. P1 is a three-line argument from trace-norm contraction, and P2 reuses the second-order bounds of Proposition 13. The alternative needs semigroup theory that neither the brief nor Mathlib has.

Confidence.

- High: some strengthening of DR-PROJ is needed. The counterexample sketch is not checked in Lean, but no argument from equal subdivisions alone is known.
- Medium-high: the subdivision form is what the author means, given "common refinements". The author may prefer the generator form; both resolve the blocker.

4.2 What perfect reconstruction means

Guess. Reconstruction is a second infinitesimal state-adaptive projective process Ψ'_τ, with its own limit Φ'_t, that runs the evolution backwards, and prediction and reconstruction undo each other on pure states, in both orders. In the language of branches, for every $t > 0$ and every pure state P:

- a predicted branch of $(\Psi_{t/n})^n$ from P has probability tending to one, and a reconstructed branch of $(\Psi'_{t/n})^n$ from the predicted state $\Phi_t P$ has probability tending to one and ends at P;
- the same with the roles of Ψ and Ψ' exchanged.

By P3 this is the same as $\Phi'_t(\Phi_t P) = P$ and $\Phi_t(\Phi'_t P) = P$ for every pure P, which is what the formalization assumes (`DrProj`). The Lean proof then stands as it is. For the resolvents $\Psi'_\tau = (I + \tau L)^{-1}$, which the brief already names; the brief may also write it as $\Psi_{-\tau}$, so that DR-PROJ covers both directions of time at once.

Why.

- It is what the brief already says. The proof needs "every pure future state reconstructs a unique pure initial state", and the remark "The finite process is not unitary" says that "in the infinitesimal limit ... prediction and reconstruction become mutually inverse". Mutually inverse means both orders.
- One order is not enough. $\Phi'_t \circ \Phi_t = I$ on pure states makes Φ_t injective there, but Proposition 17 needs it onto the pure states as well. That comes from $\Phi_t \circ \Phi'_t = I$: every pure state is the evolution of the state it reconstructs to.
- Probabilities alone say nothing about where the reconstruction ends. A branch that is certain but ends at a state other than P does not reconstruct P, so the reconstructed branch must start from the predicted state and end at the initial one.
- Physically, reconstruction is retrodiction: from the present state one infers the past state with certainty. The two sides of the guess are retrodiction of every prediction and prediction of every retrodiction.

Remark. In finite dimension reconstruction may not be needed at all. Each Ψ_τ fixes the maximally mixed state I/d, since $\sum_a P_a(I/d)P_a = I/d$, so Φ_t is affine, unital and maps pure states to pure states. By the classification of linear maps that preserve pure states, it is then conjugation by a unitary or an anti-unitary, as the constant maps onto one pure state are not unital for $d \geq 2$. This sketch is not checked in Lean. In infinite dimension there is no maximally mixed state, and an isometry that is not unitary is not excluded this way, so reconstruction is needed there, or at least nothing shown here replaces it.

Confidence.

- High: the reconstruction inverts the evolution on pure states, in both orders; the brief says "mutually inverse".
- Medium: the wording through branches. The author may prefer to state reconstruction directly through the limits, $\Phi'_t \Phi_t = \Phi_t \Phi'_t = I$ on pure states.

5 Mathematical prerequisites to add to the brief

Each item is a purely mathematical statement for Lean.

P1. **Composition from subdivisions.** Let $(\Psi_\tau)_{\tau>0}$ be maps of the density operators that are affine, and so contract the trace norm. Suppose that for every $t > 0$ the ordered products $\Psi_{\tau_k} \circ \cdots \circ \Psi_{\tau_1}$ over subdivisions of $[0, t]$ converge on every density operator in trace norm to Φ_t, as the longest interval tends to zero. Then $\Phi_{s+t} = \Phi_t \circ \Phi_s$ for all $s, t > 0$. Proof: for subdivisions A_n of $[0, t]$ and B_n of $[0, s]$, $\|A_n B_n \rho - \Phi_t \Phi_s \rho\|_1 \le \|B_n \rho - \Phi_s \rho\|_1 + \|A_n \Phi_s \rho - \Phi_t \Phi_s \rho\|_1 \to 0$, and $A_n B_n$ runs over subdivisions of $[0, s + t]$.
Lean: not formalized. Short, from `AffineOnDensityOperators.traceENorm_sub_le` and `isDensityOperator_of_tendsto_traceENorm`.

P2. **Subdivisions of the cutoff resolvents.** Let $U_t = e^{tH/(i\hbar)}$ for a self-adjoint, possibly unbounded H, with $T_t \rho = U_t \rho U_t^\dagger$, and let Ψ_τ be the cutoff resolvents (23) with M_τ decreasing in τ. Then $\Psi_{\tau_k} \circ \cdots \circ \Psi_{\tau_1} \rho \to T_t \rho$ in trace norm over every subdivision of $[0, t]$ whose longest interval tends to zero. Proof: if $P_m \rho = \rho$ for the spectral projector P_m of H on $[-m, m]$, the states $T_s \rho$ stay in the range of P_m, and once $M_{\tau_i} \ge m$ each step is the resolvent of the bounded group with generator HP_m. Telescoping with the bound (14) gives an error of at most $\sum_i 80\, m^2 \tau_i^2/\hbar^2 \le 80\, m^2\, t \max_i \tau_i/\hbar^2$. Such ρ are dense, and contraction extends the limit to every density operator.
Lean: not formalized. From `laplaceResolvent_cutoffGroup_eq`, `traceENorm_laplaceResolvent_sub_unitaryChannel_le`, `exists_isDensityOperator_mul_eq` and the telescoping bound behind `traceENorm_sub_unitaryChannel_le_of_step`.

P3. **A certain branch ends at the limit.** Let Ψ be state-adaptive projective and $Q_0, \ldots, Q_n$ a branch of n steps of Ψ with probability w. Then $\|\Psi^n Q_0 - Q_n\|_1 \le 2(1 - w)$. Hence, if $(\Psi_{t/n})^n \rho \to \Phi_t \rho$ and branches from a pure P have probability tending to one and end at states that converge to R, then $R = \Phi_t P$. With the reconstruction of the guess this gives $\Phi'_t(\Phi_t P) = P$ and $\Phi_t(\Phi'_t P) = P$, so Φ_t is a bijection of the pure states.
Lean: the bound is `traceENorm_iterate_sub_le_of_isBranch`; the bijection is `DrProj.drEv`. The step from branches to limits is not stated, but follows at once.

The proof of Theorem 18 then uses P1 for composition, P2 for the strengthened DR-UNIT ⇒ DR-PROJ, and P3 with Proposition 17 for the unitary implementation.

6 Other changes to the brief

1. **DR-PROJ.** Replace "applying the same process of duration t/n, n times, in the limit $n \to \infty$" by "applying the process over any subdivision of t, in the limit where the longest interval tends to zero". Say that $\Phi_0 = I$. Replace "the probabilities of the predicted and reconstructed state sequences tend to one" by: there is a reconstruction process, again infinitesimal and state-adaptive projective, such that from every pure state the predicted sequence and the sequence that reconstructs its starting point from its end both have probability tending to one, and the same with prediction and reconstruction exchanged.

2. **Proof of DR-UNIT ⇒ DR-PROJ.** State eqs. (20) and (24) for subdivisions, citing P2; the estimates do not change. For reconstruction, add that the limit of $(I + \tau L_\tau)^{-1}$ is T_{-t}, which inverts T_t, so both orders hold.

3. **Proof of DR-PROJ ⇒ DR-UNIT.** Replace "Perfect reconstruction gives the converse correspondence: every pure future state reconstructs a unique pure initial state" by a citation of P3. Replace the paragraph on common refinements and the remaining interval, from "The same argument gives composition" to "$\Phi_t \circ \Phi_s$ for all $t, s \geq 0$", by a citation of P1. Eq. (29) can stay as an example. Drop "Therefore $\Phi_0 = I$", which is now part of DR-PROJ.

4. **Remark "Mathematical meaning of prediction and reconstruction".** Align it with the new DR-PROJ: the limit is over subdivisions, and the reconstructed branch starts from the predicted state and ends at the initial one. The clause that the predicted states converge can go, since P3 gives it.

When the blockers close. Both brief issues close once the brief adopts these readings, or the alternatives. In Lean, `IsProjectiveLimit` then takes the limit over subdivisions. With P1 and P2, `IsProjectiveLimit.isEvolution` no longer assumes composition for irrational ratios, and $\Phi_0 = I$ becomes part of the definition, so DR-UNIT ⇔ DR-PROJ holds without the extra clauses of `drUnit_iff_drProj_and_map_zero_and_map_add`. `DrProj` already matches the guess on reconstruction; adding the branch form and P3 is short.